

Verification of FastAPI - Performance and asynchronous requests

Create **fastapi_async.py** file:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ uvicorn fastapi_async:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api/performance_and_async_requests']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [2368] using StatReload
INFO: Started server process [2377]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open URL <http://127.0.0.1:8000/docs>

FastAPI 0.1.0 OAS 3.1
openapi.json

default ^

GET	/sync	Get Sync	▼
GET	/async	Get Async	▼

Create **test_requests.py**.

Open new terminal to run following command:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ python3 test_requests.py
making 20 requests on the '/sync' endpoint ...
took 10.052160823345185 seconds
making 20 requests on the '/async' endpoint
took 0.0879289150238037 seconds
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$
```

Modify **fastapi_async.py** so that async endpoint waits the 10 seconds (part 2)

Restart Uvicorn:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ uvicorn fastapi_async:api --reload
INFO: Will watch for changes in these directories: ['/home/ubuntu/datascientest-data-engineering/fast-api/performance_and_async_requests']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [4248] using StatReload
INFO: Started server process [4250]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Rerun load test:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ python3 test_requests.py
making 20 requests on the '/sync' endpoint ...
took 10.087979292869568 seconds
making 20 requests on the '/async' endpoint
took 10.058924233913421 seconds
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$
```

Some curl sanity checks:

```
● ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ curl -i http://127.0.0.1:8000/sync
HTTP/1.1 200 OK
date: Thu, 19 Jun 2025 13:29:23 GMT
server: uvicorn
content-length: 25
content-type: application/json

● {"message":"synchronous"}ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$ curl -i http://127.0.0.1:8000/async
HTTP/1.1 200 OK
date: Thu, 19 Jun 2025 13:29:41 GMT
server: uvicorn
content-length: 26
content-type: application/json

○ ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/fast-api/performance_and_async_requests$
```

Open <http://127.0.0.1:8000/sync>

```
▼ {
  "message": "synchronous"
}
Raw  Parsed
```

Open <http://127.0.0.1:8000/async>

```
▼ {
  "message": "asynchronous"
}
Raw  Parsed
```